# TEST 10—FORM CODE 6G

**SECTION 1** Time—30 minutes, 35 Questions

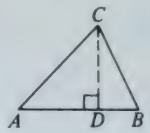
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius $r$: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

| | |
|---|---|
| $=$ is equal to | $\leqq$ is less than or equal to |
| $\neq$ is unequal to | $\geqq$ is greater than or equal to |
| $<$ is less than | $\parallel$ is parallel to |
| $>$ is greater than | $\perp$ is perpendicular to |




Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

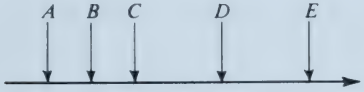
(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. Bob makes 720 telephone calls each year. What is the average (arithmetic mean) number of calls that Bob makes per month?

(A) 60
(B) 70
(C) 80
(D) 100
(E) 120




2. On the number line above, if the numbers 3, −1, 5, 0, and 1 are indicated by the arrows, which arrow indicates 0?

(A) $A$ (B) $B$ (C) $C$ (D) $D$ (E) $E$

3. If $3 + x = 2x + 1$, then $x =$

(A) $\frac{2}{3}$
(B) 1
(C) $\frac{4}{3}$
(D) 2
(E) 4

4. $\sqrt{25 \times 64} =$

(A) 1,600 (B) 400 (C) 80
(D) 40 (E) 13

5. A beach umbrella was rented each day for one week at the rate of \$8 per day on days that were sunny and \$3 per day on days that were not sunny. If the total rental charge for that week was \$46, what was the number of sunny days in that week?

(A) 6
(B) 5
(C) 4
(D) 3
(E) 2

6. If $\frac{N \times N}{N + N} = 1$, then $N =$

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2 (E) 4

7. A square and an equilateral triangle have sides of lengths $x$ and $y$, respectively. The sum of their perimeters can be represented by

(A) $x + y$ (B) $3x + 4y$ (C) $4x + 3y$
(D) $7(x + y)$ (E) $x^2 + y^2$

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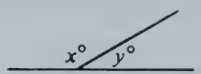
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Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

**AN E RESPONSE WILL NOT BE SCORED.**

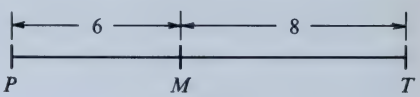
EXAMPLES

| | Column A | Column B | Answers |
|---|---|---|---|
| E1. | $2 \times 6$ | $2 + 6$ | ● Ⓑ Ⓒ Ⓓ Ⓔ |
| E2. |  $180 - x$ | $y$ | Ⓐ Ⓑ ● Ⓓ Ⓔ |
| E3. | $p - q$ | $q - p$ | Ⓐ Ⓑ Ⓒ ● Ⓔ |

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as $x$, $n$, and $k$ stand for real numbers.

| | Column A | Column B |
|---|---|---|
| 8. | The least positive integer divisible by 2, 3, and 4 | 24 |
| | $n > 0$ | |
| 9. | 40% of $n$ | 20% of $2n$ |
| | Parallel lines $\ell_1$ and $\ell_2$ are 2 inches apart. $P$ is a point on $\ell_1$ and $Q$ is a point on $\ell_2$. | |
| 10. | Length of $PQ$ | 3 inches |
| 11. | The sum of 3 consecutive integers if the middle integer is 8 | The sum of 4 consecutive integers if the least integer is 5 |
| | $x = 3$ | |
| 12. | $(4x)^5$ | $4x^5$ |
| | $k + k + k + k = 8$<br>$3(m + 2m) = 18$ | |
| 13. | $k$ | $m$ |

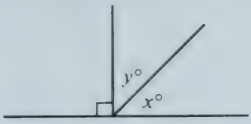
| | Column A | Column B |
|---|---|---|
| | $x + y + z = 6$ | |
| 14. | $2x + 2y + 2z$ | 14 |
| | $x < y$ | |
| 15. | $3x$ | $2y$ |
| |  Point $R$ (not shown) is between $P$ and $T$ on line segment $PT$. The length of $RT$ is 9. | |
| 16. | Length of $RM$ | 2 |
| | $x$ and $y$ are different even integers greater than 2. | |
| 17. | The number of odd integers between $x$ and $y$ | The number of prime numbers between $x$ and $y$ |
| | $\frac{2}{3} = \frac{4}{x} = \frac{8}{y}$ | |
| 18. | $x + y$ | 18 |

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**SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS**

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

**AN E RESPONSE WILL NOT BE SCORED.**

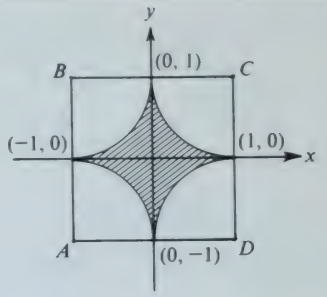
| | Column A | Column B |
|---|---|---|
| |   $y > x$ | |
| 19. | $x$ | 45 |
| 20. | $\dfrac{1}{2-\frac{1}{2}}$ | $\dfrac{1}{2-\dfrac{1}{2-\frac{1}{2}}}$ |
| | The three angles of a triangle are $x°$, $x°$, and $\frac{x°}{2}$. | |
| 21. | $x$ | 60 |
| | $x = y \neq 0$ | |
| 22. | $xy$ | 0 |

Questions 23-25 refer to the following definition of $\widehat{a, b, c}$ for all real numbers $a$, $b$, and $c$.

$$\widehat{a, b, c} = (a - b)(b - c)$$

| | | |
|---|---|---|
| 23. | $\widehat{1, 1, x}$ | $\widehat{2, 2, x}$ |
| 24. | $\widehat{0, 1, x}$ | $\widehat{1, 0, x}$ |
| 25. | $\widehat{x, 2, 1}$ | $x - 3$ |

| | Column A | Column B |
|---|---|---|
| 26. | $2^{10} - 2^9$ | $2^9$ |




Quadrilateral $ABCD$ is a square with sides parallel to the axes. The shaded region is bounded by arcs of circles with radii 1 and centers $A$, $B$, $C$, and $D$.

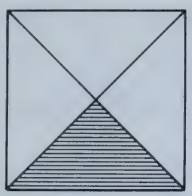
| | | |
|---|---|---|
| 27. | The area of the shaded region | 1 |

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Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. Suppose that angles of geometric figures were measured in "bleeps" instead of in degrees. If the sum of the measures of the angles of a triangle were 12 bleeps, what would be the sum of the measures of the angles of a rectangle, in bleeps?

(A) 6
(B) 12
(C) 16
(D) 24
(E) 36



29. If the shaded part of the square above has area 16, what is the perimeter of the square?

(A) 4 (B) 8 (C) 16 (D) 32 (E) 64

30. On a number line, if point $A$ has coordinate $3\frac{1}{2}$ and point $B$ has coordinate $-1\frac{1}{2}$, what is the coordinate of the point equidistant from $A$ and $B$?

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) 1 (D) $1\frac{1}{2}$ (E) 2

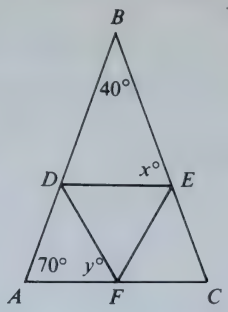
31. A total of \$24.00 is to be divided among three children in amounts proportional to their ages. If the children's ages are 9, 12, and 15 years, how much money will the youngest child receive?

(A) \$2.00 (B) \$3.00 (C) \$4.00
(D) \$5.00 (E) \$6.00

32. If the product of two positive integers $r$ and $s$ is 7,240, which of the following must be true?

I. Both $r$ and $s$ are even numbers.
II. Either $r$ or $s$ is a multiple of 10.
III. Either $r$ or $s$ is a multiple of 5.

(A) None (B) I only (C) II only
(D) III only (E) I, II, and III




33. In the figure above, $\triangle DEF$ is equilateral. If $DE \parallel AC$, then $x - y =$

(A) 0 (B) 5 (C) 10 (D) 15 (E) 20

34. If $x$ multiplied by $\frac{11}{2}$ gives the same result as $x$ subtracted from $\frac{11}{2}$, what is the value of $x$?

(A) $\frac{11}{13}$ (B) 1 (C) $\frac{11}{9}$ (D) $\frac{11}{3}$ (E) $\frac{11}{2}$

35. If Sue is not on the Planning Committee, from which of the following statements can it be determined whether Sue is on the Rules Committee?

(A) Everyone on the Planning Committee is on the Rules Committee.
(B) Everyone on the Rules Committee is on the Planning Committee.
(C) Some people not on the Planning Committee are also not on the Rules Committee.
(D) Anyone not on the Rules Committee is not on the Planning Committee.
(E) No one is on both the Planning Committee and the Rules Committee.

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

**SECTION 2** Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. FRISKY: (A) tiny (B) perfect (C) alone (D) sluggish (E) flexible

2. PROPEL: (A) indent (B) expose (C) dismount (D) hold back (E) put together

3. SQUABBLE: (A) permit (B) agree (C) avoid (D) add to (E) gloss over

4. KINDLE: (A) distribute (B) extinguish (C) make poisonous (D) cause rigidity (E) mix vigorously

5. AWESOME: (A) unfamiliar (B) uniform (C) insignificant (D) inconvenient (E) misleading

6. CONVERGE: (A) remain silent (B) grow weaker (C) despair (D) resume (E) separate

7. PLAUSIBLE: (A) symmetrical (B) unbelievable (C) randomly chosen (D) unclearly stated (E) extremely long

8. INSTIGATION: (A) inhibition (B) insulation (C) intuition (D) abstraction (E) jubilation

9. CONTENTIOUS:
(A) eager to please
(B) ready to leave
(C) anxious to intervene
(D) afraid of danger
(E) suspicious of flattery

10. DERISION: (A) conclusion (B) applause (C) safety (D) gradual decline (E) sudden return

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. The ---- of trees in East Africa has caused the number of native antelopes to ---- sharply because they can live only where the forest is most dense.
(A) planting..decline
(B) destruction..grow
(C) felling..decrease
(D) abundance..dip
(E) protection..drop

12. The speaker stated that progress in the civil rights movement would be ---- unless organizations move beyond ---- the problems of the times to developing strategies for solving them.
(A) comprehensive..sustaining
(B) minimal..defining
(C) painstaking..rejuvenating
(D) trivial..eradicating
(E) overwhelming..analyzing

13. Even though the task of editing the series seems to become increasingly ---- with each volume, Donna feels compelled to ---- the project until others can take over the responsibility.
(A) creative..enlarge
(B) inspiring..publicize
(C) feasible..malign
(D) monotonous..relinquish
(E) demanding..continue

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14. The committee's report was entirely ----; those who ---- it did so with the deliberate intent of damaging my reputation.
    (A) erroneous..expunged
    (B) legitimate..perused
    (C) infallible..endorsed
    (D) spontaneous..compiled
    (E) malicious..circulated

15. As a scientific work, the paper is flawed by the inclusion of many ---- details that are not sufficiently ---- to the overall thesis.
    (A) picayune..alien
    (B) verifiable..contradictory
    (C) ancillary..germane
    (D) integral..incidental
    (E) inconspicuous..detrimental

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. VEAL:CALF :: (A) egg:chicken (B) lamb:goat (C) milk:cow (D) pork:pig (E) duck:goose

17. RESTRAIN:IMMOBILIZE ::
    (A) hinder:prevent
    (B) forbid:defy
    (C) camouflage:substitute
    (D) excuse:resent
    (E) store:misplace

18. SOLDIER:INFANTRY ::
    (A) salesperson:convention
    (B) musician:rhythm
    (C) athlete:team
    (D) sprinter:marathon
    (E) violinist:solo

19. SENTINEL:PROTECTION ::
    (A) adversary:relief
    (B) arbitrator:domination
    (C) interpreter:translation
    (D) idler:reminiscence
    (E) nominee:election

20. SHARD:POTTERY :: (A) tack:metal (B) chip:stone (C) slab:meat (D) varnish:wood (E) flask:water

21. DISCONTENTED:MOROSE ::
    (A) successful:vain
    (B) proud:arrogant
    (C) nonchalant:tense
    (D) helpful:useful
    (E) jovial:unpopular

22. UNPRECEDENTED:PREVIOUS OCCURRENCE ::
    (A) obvious:conformity
    (B) peerless:equal
    (C) untimely:irregularity
    (D) unique:exception
    (E) erratic:mobility

23. MOUNTAIN:RANGE :: (A) lake:ocean (B) glacier:snow (C) island:archipelago (D) sand:dune (E) surf:beach

24. DULCET:HEARING ::
    (A) wishful:realizing (B) savory:tasting (C) sincere:speaking (D) sobbing:weeping (E) inquisitive:learning

25. HIDEBOUND:INFLEXIBILITY ::
    (A) suspicious:guilt
    (B) audacious:reserve
    (C) compassionate:sympathy
    (D) successful:admiration
    (E) extravagant:elegance

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Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

In most of the truest and most fully satisfying fiction written in the United States after the Second World War, the newborn or self-sustaining or orphaned hero is plunged again and again, for his or her own good and for ours, into the spurious, disruptive rituals of the actual world. We may mention especially *Invisible Man*, by Ralph Ellison; J. D. Salinger's *The Catcher in the Rye*; and *The Adventures of Augie March*, by Saul Bellow—novels in which the heroes are willing, with marvelously inadequate equipment, to take on as much of the world as is available to them, without ever fully submitting to any of the world's determining categories. Ellison's hero is a nameless Black, Salinger's an unstable adolescent, Bellow's a rebellious Chicago Jew. But they share in their common aloneness an odd aura of moral priority over the waiting world. Each of them struggles tirelessly, sometimes unwittingly, and often absurdly, to realize the full potentialities of the classic type each represents—the Emersonian "simple genuine self against the whole world." These characters are clearly indebted to the great innocents of nineteenth-century American fiction, but their creators are also able to engender from within their work the hopeful and vulnerable sense of life that makes experience and so makes narrative action possible. To test that sense of life by irony and drama they must create it from within, for they can scarcely find it any longer in the historic world about them.

26. This passage is primarily concerned with
    (A) defining the nature of the hero in certain contemporary fiction
    (B) criticizing the lack of originality in popular twentieth-century fiction
    (C) giving examples of the instability of modern society
    (D) prescribing the direction that fiction should take
    (E) satirizing the inadequate heroes of recent fiction

27. The author suggests that all of the characters discussed share the quality of being
    (A) sophisticated (B) defeatist (C) naïve (D) impulsive (E) intellectual

28. The author's attitude toward the fictional characters mentioned seems to be
    (A) neutral (B) contemptuous (C) despairing (D) reverent (E) admiring

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The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

Both living and nonliving things interact with their environments across their surfaces. The nature of the surface greatly affects the kinds of interactions that can occur. As a rule, a smooth surface has less functional surface area per square meter than a rough or wrinkled one, and a larger ratio of surface area to volume will permit more rapid interaction of materials. Finely granulated sugar, for instance, dissolves more rapidly in a cup of coffee than a lump of sugar does.

During the course of evolution, modification of surface area has been an important adaptation of organisms to the environment. The wall of the human small intestine, for example, is layered with microvilli that provide greater surface area for absorption of food. Similarly, in plant roots epidermal cells have extensions (root hairs) that increase a plant's ability to take in water and minerals from the soil.

Unlike the interchanges that occur at the surfaces of most nonliving substances (such as the sugar cube), exchanges of materials in living cells occur through cell membranes that are selectively permeable. Cell membranes regulate the movement of most substances in and out of the cell, allowing passage of some substances and blocking others.

Cell membranes, however, appear unable to regulate the movement of water. Over the millenia, organisms have evolved many strategies to conserve water. Cactus plants have adapted to their hot, dry environment by developing thickened leaves with relatively reduced surface area. In areas where the lack of water is not so severe, most plants have leaves with smaller volumes and larger surface areas. Broad-leaved plants such as beech and maple trees lose water rapidly through transpiration and thrive where moisture is abundant and temperatures are moderate.

29. This passage primarily discusses
    (A) the difference between the surfaces of living things and the surfaces of nonliving things
    (B) how the roughness of an object's surface affects its permeability
    (C) in what manner various living things have adapted to their environment
    (D) why cell membranes are unable to control the absorption and release of water
    (E) how the surface area of living and nonliving things affects their interaction with their environment

30. In lines 13-18, the author mentions microvilli in human intestines and root hairs on plant roots as illustrations of adaptations that
    (A) decrease the cells' dependence on water
    (B) increase the absorptive ability of the organism
    (C) retard the exchange of material between the cells and the environment
    (D) maintain the flexibility of the cellular walls
    (E) decrease the surface area of the cellular walls

31. According to information in the passage, "most substances" (lines 23-24) would include which of the following?
    I. Food
    II. Water
    III. Minerals
    (A) II only (B) III only (C) I and II only (D) I and III only (E) I, II, and III

32. It can be inferred from the passage that the surface of which of the following is LEAST conducive to rapid interaction of materials?
    (A) A prune (B) A sponge (C) A piece of cake (D) A balloon (E) A tree root

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Many years later, he was to remember that distant afternoon with his father, Arcadio. The world was so recent that many things lacked names, and in order to indicate them it was necessary to point. Every year during the month of March a family of gypsies would set up their tents near the village, and with a great uproar of pipes and kettledrums they would display new inventions. First they brought the magnet. A heavy man with an untamed beard and sparrow hands, who introduced himself as Melquiades, put on a bold public demonstration of what he himself called the eighth wonder of the learned alchemists of Macedonia. He went from house to house dragging two metal ingots and everybody was amazed to see pots, pans, tongs, and braziers tumble down from their places and beams creak from the desperation of nails and screws trying to emerge. Arcadio, whose unbridled imagination always went beyond the genius of nature and even beyond miracles and magic, thought that it would be possible to make use of that useless invention to extract gold from the bowels of the earth. Melquiades, who was an honest man, warned him: "It won't work for that." But Arcadio at that time did not believe in the honesty of Melquiades, so he traded his mule and a pair of goats for the two magnetized ingots. He explored every inch of the region, even the riverbed, dragging the two iron ingots along and reciting Melquiades' incantation aloud. The only thing he succeeded in doing was unearthing a rusted suit of fifteenth-century armor.

33. The passage focuses primarily on
    (A) a man's desire to relive the past
    (B) a childhood incident as it is later recalled
    (C) an account of a day in the life of a traveler
    (D) an invention and its impact on a rural village
    (E) the poverty of a man's childhood environment

34. It can be inferred that the speaker refers to things lacking names (line 3) in order to suggest the
    (A) whimsical quality of the relationship between the speaker and his father
    (B) importance of technological advances in remote areas of the world
    (C) overwhelming interest in the gypsies' visit
    (D) recency of the establishment of the village
    (E) freshness of childhood perceptions

35. Which of the following best expresses Arcadio's reaction to the metal ingots?
    (A) He was frightened by their magical power.
    (B) He suspected all along that they were inherently worthless.
    (C) He resented the way they were used to exploit the villagers.
    (D) He remained aloof from all the commotion they caused.
    (E) He was determined to find a profitable application for them.

---

The study of ethics is perhaps most commonly conceived as being concerned with the questions, "What sort of actions ought people to perform?" and "What sort of actions ought they to avoid?" As a result, the study of ethics is usually regarded as the practical study, to which all others may be opposed as theoretical; the good and the true are spoken of as independent kingdoms, the former belonging to ethics, the latter to science.

This view, however, is doubly defective. In the first place, it overlooks the fact that the object of ethics is to discover true propositions about virtuous and vicious conduct and that these are just as much a part of truth as true propositions about oxygen or the multiplication table. The aim is not practice, but propositions about practice; and propositions about practice are not themselves practical, any more than propositions about gases are gaseous. One might as well maintain that botany is vegetable or zoology animal. Thus the study of ethics is not something outside science and coordinate with it: it is one among sciences.

In the second place, the view in question unduly limits the province of ethics. When we are told that actions of certain kinds ought to be performed or avoided, as for example that we ought to speak the truth, we may always legitimately ask for a reason, and this reason will be concerned not only with the actions themselves, but also with the goodness or badness of the consequences likely to follow from such actions. We shall be told that truth-speaking generates mutual confidence and facilitates the dispatch of business, and hence increases the wealth of the society that practices it. If we ask why we should aim at increasing mutual confidence or increasing the wealth of society, we may be told that obviously these things are good or that they lead to happiness and happiness is good. If we still ask why, the ordinary person will probably feel irritation and will reply that

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no one knows.

However, it is the business of the student of ethics to ask for reasons as long as reasons can legitimately be demanded and to register the propositions which give the most ultimate reasons that are attainable. Since a proposition can be proven only by means of other propositions, it is obvious that not all propositions can be proven, for proofs can begin only by assuming something. The first step in ethics, therefore, is to be quite clear as to what we assume when we conceive of "good" and "bad." Only then can we begin to formulate propositions about conduct and ask how right conduct is related to the production of good and the avoidance of evil.

36. In the passage, the author is primarily concerned with
    (A) proving that ethics is a practical study
    (B) disputing the claim that ethics is a scientific discipline
    (C) challenging common notions about the province of ethics
    (D) proving the basic assumptions of ethics
    (E) defining what is meant by good and evil

37. The author most probably includes the questions in lines 3-4 in order to do which of the following?
    (A) Emphasize the narrowness of ethical concerns
    (B) Provide an example of a practical and a theoretical issue
    (C) Allude to recent changes in the study of ethics
    (D) Emphasize the conflicting objectives of ethics
    (E) Illustrate what ethics is usually thought to encompass

38. The "view in question" (line 23) refers to the view that
    (A) propositions about actions should be evaluated in terms of the goodness or badness of their consequences
    (B) propositions about actions are based on unproven assumptions
    (C) ethics deals exclusively with the good and science exclusively with the true
    (D) ethics deals with the discovery of true propositions about actions
    (E) ethics is not independent of science but merely one of its disciplines

39. According to the author, some propositions cannot be proven because
    (A) the number of possible true propositions is unlimited
    (B) propositions about conduct require knowledge of human motives
    (C) the ultimate consequences of a proposition cannot always be foreseen
    (D) the truth of propositions can be determined only through other propositions
    (E) the goodness or badness of some propositions is self-evident

40. The author most probably refers to the increase in wealth promoted by the practice of truth-speaking (lines 31-34) in order to
    (A) prove that good actions are those that benefit a large group of people
    (B) suggest that the process of evaluating the consequences of an action can never end
    (C) imply that actions that appear to be good may have questionable consequences
    (D) show how an action that is neither bad nor good can have beneficial consequences
    (E) illustrate how one can begin to evaluate an action in terms of its consequences

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

**SECTION 5** Time—30 minutes, 25 Questions

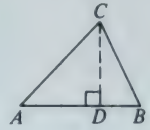
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius $r$: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

| | |
|---|---|
| $=$ is equal to | $\leqq$ is less than or equal to |
| $\neq$ is unequal to | $\geqq$ is greater than or equal to |
| $<$ is less than | $\parallel$ is parallel to |
| $>$ is greater than | $\perp$ is perpendicular to |




Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

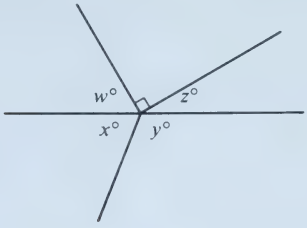
Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $a = -1$ and $b = 2$, then $a^2 + b =$

(A) $-1$
(B) 0
(C) 1
(D) 3
(E) 4

2. The cost to construct a fence 10 meters long is \$350. At this rate, what is the cost to construct a fence 15 meters long?

(A) \$400
(B) \$420
(C) \$500
(D) \$525
(E) \$1,750




3. In the figure above, $w + x + y + z =$

(A) 360
(B) 300
(C) 270
(D) 240
(E) 180

4. Each of the following two-digit numbers has one digit covered. Which of these five numbers is the only possible multiple of 12?

(A) ▨9
(B) ▨5
(C) ▨3
(D) 3▨
(E) 5▨

5. When $2x$ is multiplied by $3x$, the product is

(A) 5
(B) $5x$
(C) $6x$
(D) $5x^2$
(E) $6x^2$

6. If the fraction $\frac{x}{7}$ is between 2 and 3, which of the following numbers is a possible value of $x$?

(A) 10
(B) 12
(C) 16
(D) 22
(E) 26

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7. $(6 \div 3) \div 5 =$

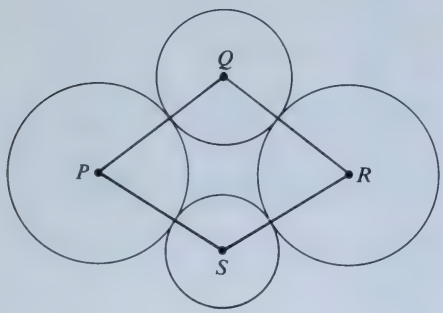
(A) 0.4 (B) 0.6 (C) 1.0
(D) 2.5 (E) 10.0

8. A rectangle with length 12 and width 8 has an area equal to the sum of the areas of how many squares each with side of length 4 ?

(A) 3
(B) 6
(C) 8
(D) 24
(E) 96

9. Mary ran 6 kilometers in 30 minutes. What was her average speed in kilometers per hour?

(A) 5
(B) 10
(C) 12
(D) 15
(E) 30

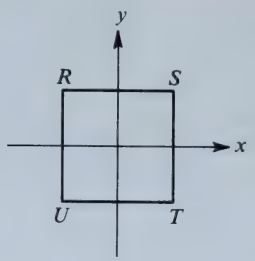



10. In the figure above, the circles with centers $P$ and $R$ have radius 8, the circle with center $Q$ has radius 6, and the circle with center $S$ has radius 5. If the circles are tangent as shown, what is the perimeter of quadrilateral $PQRS$?

(A) 27 (B) 38 (C) 44 (D) 50 (E) 54

11. If a department store offers an item originally priced at \$8.00 at a reduced price of \$6.00, by what percent is the original price reduced?

(A) 10%
(B) $12\frac{1}{2}\%$
(C) 20%
(D) 25%
(E) $33\frac{1}{3}\%$




12. In the figure above, the center of square $RSTU$ is at the origin and the sides of the square are parallel to the $x$- and $y$-axes. If the square is rotated through 180° about the $y$-axis and then rotated through 180° about the $x$-axis, the vertices will be in which of the following positions?

(A) $T$ $U$
$S$ $R$

(B) $T$ $S$
$U$ $R$

(C) $R$ $U$
$S$ $T$

(D) $U$ $R$
$T$ $S$

(E) $S$ $T$
$U$ $R$

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13. The number 0.253 is how much greater than $\frac{1}{4}$?

(A) $\frac{1}{3}$ (B) $\frac{3}{10}$ (C) $\frac{3}{25}$

(D) $\frac{3}{100}$ (E) $\frac{3}{1{,}000}$

---

14. Segments $AB$, $CD$, and $EF$ lie in the same plane. If $AB$ and $CD$ are perpendicular bisectors of each other and if $CD$ is perpendicular to and bisects $EF$ at point $D$, which of the following statements must be true?

I. Segment $AB$ is parallel to segment $EF$.
II. Segment $AB$ is a perpendicular bisector of segment $EF$.
III. Segment $AB$ and segment $EF$ have the same length.

(A) I only (B) II only (C) III only
(D) I and III (E) II and III

Questions 15-17 refer to the following definition.

For any integer $n$, define $\boxed{n}$ by the equation
$\boxed{n} = (n+1)n - (n-1)(n-2)$.

15. $\boxed{5} =$

(A) 18
(B) 16
(C) 14
(D) 12
(E) 10

---

16. If $\boxed{n} = 30$, then $n =$

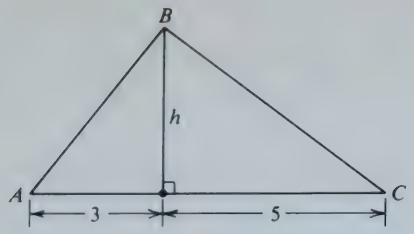
(A) 6
(B) 7
(C) 8
(D) 9
(E) 10

---

17. $\boxed{n+1} - \boxed{n-1} =$

(A) 1
(B) 2
(C) 3
(D) 4
(E) 8

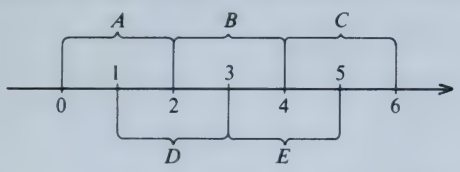
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Note: Figure not drawn to scale.

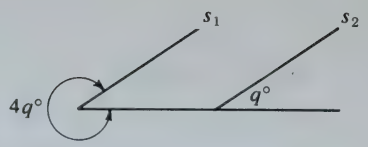
18. If the area of $\triangle ABC$ above is 30, what is the value of $h$?

(A) 3.75
(B) 6
(C) 7.5
(D) 10
(E) It cannot be determined from the information given.




19. A portion of a number line has intervals labeled as shown in the figure above. If a number $x$ is in interval $A$ and a number $y$ is in interval $D$, then the number $\frac{1}{2}(x+y+1)$ must be in interval

(A) $A$ (B) $B$ (C) $C$ (D) $D$ (E) $E$




Note: Figure not drawn to scale.

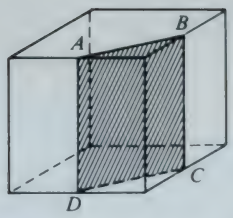
20. If $s_1 \parallel s_2$ in the figure above, what is the value of $q$?

(A) 36
(B) 60
(C) 72
(D) 90
(E) 120

21. A piece of wire 0.002 meter long is cut off from a wire 0.2 meter long. The remaining piece is how many times as long as the piece that was cut off?

(A) 40
(B) 99
(C) 100
(D) 198
(E) 1,000

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22. In the figure above, $A$, $B$, $C$, and $D$ are midpoints of four edges of the cube. If the length of an edge of the cube is $x$, what is the area of rectangle $ABCD$?

(A) $\frac{x^2}{2}$

(B) $x^2$

(C) $\frac{x^2\sqrt{2}}{2}$

(D) $x^2\sqrt{2}$

(E) $2x^2$

23. If $\frac{1}{2}n + 2 = A$, then $2n + 2 =$

(A) $2A$
(B) $2A - 2$
(C) $4A$
(D) $4A - 2$
(E) $4A - 6$

24. If the present time is $x$ minutes past $y$ o'clock, where $0 \leq x < 60$ and $y$ is a positive integer less than 12, in how many seconds will it be $(y + 1)$ o'clock?

(A) $1 - x$ (B) $60 - x$ (C) $60(1 - x)$
(D) $60(60 - x)$ (E) $60x$

25. If the average (arithmetic mean) of ten consecutive integers arranged in increasing order is $15\frac{1}{2}$, what is the average of the first five of these integers?

(A) $15\frac{1}{2}$

(B) 13

(C) $10\frac{1}{2}$

(D) 9

(E) $7\frac{3}{4}$

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

**SECTION 6** Time—30 minutes 45 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. INFLATE: (A) fail (B) deny (C) evade (D) expose (E) flatten

2. MISHAP:
   (A) fortunate event
   (B) thorough understanding
   (C) useful object
   (D) wealthy person
   (E) sincere apology

3. LENIENT: (A) stern (B) gross (C) superior (D) unlucky (E) mature

4. SUPPRESS: (A) reveal (B) await (C) receive (D) repent (E) influence

5. DIEHARD: (A) heartbreaker (B) pushover (C) simpleton (D) spoilsport (E) scatterbrain

6. UNDERMINE: (A) entangle (B) parch (C) overwork (D) enter (E) support

7. ZANY: (A) active (B) content (C) generous (D) reasonable (E) talkative

8. REPUGNANT: (A) orderly (B) appealing (C) languorous (D) haughty (E) defensive

9. INEPTNESS: (A) importance (B) eagerness (C) heaviness (D) expertise (E) novelty

10. CLANDESTINE: (A) swift (B) open (C) odorless (D) reassuring (E) conceivable

11. IMPROVIDENCE: (A) foresight (B) sentiment (C) courage (D) eloquence (E) authority

12. TRANSIENT: (A) enduring (B) indisposed (C) endangered (D) internal (E) scornful

13. CENSORIOUS: (A) adaptable (B) attentive (C) complimentary (D) sluggish (E) suspicious

14. ROTUND: (A) immoral (B) placid (C) gaunt (D) astute (E) solitary

15. CURSORY: (A) meticulous (B) approving (C) educated (D) confusing (E) delayed

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Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate
(C) sophisticated..amateur (D) risqué..crude
(E) perfect..spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Although the sociologist thought she would ---- the explanation with her charts and graphs, they actually caused additional ----.

(A) magnify..details
(B) clarify..benefits
(C) simplify..complications
(D) deprecate..problems
(E) illustrate..particulars

17. By relying on deductive reasoning, mathematicians have obtained results that would be very difficult or even ---- to obtain by other methods.

(A) essential (B) effortless (C) logical
(D) desirable (E) impossible

18. Although Helena's life was both ---- and painful, it was not without ----, for she had accomplished many of her goals.

(A) defiant..strife
(B) tragic..frustration
(C) gratifying..reward
(D) arduous..satisfaction
(E) purposeful..ambition

19. The economic landscape of the 1920's was overrun with amateur firefighters, each clutching what was believed to be a magic formula for ---- the spreading fire of inflation.

(A) fanning (B) inciting (C) quenching
(D) refueling (E) purifying

20. In the fifteenth century, ancient authority and empirical investigation were ----; not until the seventeenth century was a system created by which the two might be reconciled.

(A) unorthodox (B) incompatible
(C) inadequate (D) unintelligible
(E) interchangeable

21. The test of truth is not ----, for we have often felt firmly convinced of many things that were not so.

(A) implication (B) certitude
(C) originality (D) impartiality
(E) moderation

22. Although there is not a great deal we can do to prevent storage of information about ourselves in computer data banks, we are not entirely ----.

(A) indispensable (B) anonymous
(C) neglectful (D) helpless
(E) remorseless

23. Although Chester has his ----, he is no one's ----, for he has his own style of thinking and it is not derivative.

(A) detractors..critic
(B) followers..exemplar
(C) competitors..rival
(D) forerunners..compatriot
(E) predecessors..disciple

24. The tone of this autobiography becomes ---- as the author ---- the applause, the accolades, and the prizes that his labors have earned him.

(A) sullen..revels in
(B) smug..basks in
(C) boastful..atones for
(D) humble..exaggerates
(E) irresolute..relishes

25. The argument that science periodicals neither acknowledge nor ---- fraud in the scientific community is belied by this leading scientific journal's investigation of fabricated research and corrupt methodology.

(A) indulge (B) exploit (C) allow
(D) police (E) perpetrate

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Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

One of the most serious violations of civil liberties in the United States was the treatment of Japanese-Americans during the Second World War. In 1942 the army removed all persons of Japanese descent from coastal areas of three West Coast states and moved them inland to "relocation camps" in which conditions were far from satisfactory. The justification given for this drastic step was that it was necessary to prevent spy activities and sabotage in an essential military zone. But two-thirds of those detained were American citizens. In 1944 the Supreme Court finally ruled on the legal issues involved, holding that the mass evacuation had been legal under the war powers of the President, but that American citizens against whom no charge had been filed and whose loyalty was not questioned could not be detained after removal from the military zone. In the meantime, the War Department repealed its ban against the return of loyal citizens to the coast, but certain elements of the public in the West bitterly opposed the return of the Japanese-Americans and no less than fifty-nine acts of violence were committed in attempts to terrorize them. Although the majority of the people on the West Coast condemned such tactics, many of the Japanese-Americans decided to find new homes in parts of the country where they would encounter less prejudice.

In Hawaii between thirty and forty percent of the population was of Japanese ancestry; hence, it was not possible to remove those persons to the mainland, as was actually suggested, without completely disrupting the economic life of the islands. Therefore, Japanese-Americans in Hawaii were not relocated, but the political rights of all citizens, regardless of their race, color, or nationality, were almost completely abolished during the war. Military law replaced civilian government, and army officers discharged the duties normally assigned to judges and police magistrates. Regulations that might have had meaning right after the 1941 Japanese bombing of Pearl Harbor in Hawaii were enforced with military ruthlessness long after the reasons for their adoption had disappeared.

In spite of the violence and wholesale discrimination against Japanese-Americans, thousands of Hawaiian and mainland Americans of Japanese ancestry served with the United States armed services. Japanese-American battle units had outstanding records in the European theater of war to which they were assigned. No Japanese-American was convicted of either sabotage or espionage on the mainland or in Hawaii before or during the war.

26. It can be inferred from the passage that, according to the 1944 Supreme Court decision, it was legal for the United States government to do which of the following?

    I. Remove all people of Japanese ancestry from coastal areas of three West Coast states
    II. Detain in relocation camps people of Japanese ancestry who were not American citizens
    III. Detain in relocation camps American citizens of Japanese ancestry whose loyalty to the United States was questioned

    (A) III only (B) I and II only
    (C) I and III only (D) II and III only
    (E) I, II, and III

27. The author would be most likely to agree with which of the following statements about the enforcement of strict military regulations in Hawaii?

    (A) It was justified as long as a state of war existed.
    (B) It was reasonable immediately after Pearl Harbor was bombed.
    (C) It was defensible if the rules were applied to only certain elements of the population.
    (D) It was necessary because Hawaii was economically important.
    (E) It was unavoidable because Hawaii was so politically unstable.

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28. According to the passage, Japanese-Americans were not removed from Hawaii during the Second World War because
    (A) the military authorities on Hawaii were able to detain them in relocation camps
    (B) Hawaii was under attack from the Japanese
    (C) there was little resentment against them in Hawaii
    (D) they were important to the economic life of Hawaii
    (E) the distance between Hawaii and the relocation camps on the mainland was too great

29. The passage suggests that which of the following statements about American soldiers of Japanese ancestry during the Second World War is true?
    (A) They were not involved in combat.
    (B) They were under stricter military regulations than other soldiers.
    (C) They were in segregated battalions.
    (D) They could enlist but they were not drafted.
    (E) They were kept in relocation camps unless they volunteered for overseas service.

30. Which of the following would be the best title for the passage?
    (A) The Role of Japanese-Americans in Hawaii and on the West Coast of the United States
    (B) Hostility Toward Japanese-Americans During the Second World War
    (C) The Second World War: The Japanese-American Response
    (D) American Attitudes During the Second World War
    (E) Military Intervention in Hawaii During the Second World War

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In February 1902 Lee De Forest wrote, "I shall move all heaven and earth to put in at once a broad fundamental patent on telephony without wires by hertzian waves." As a graduate student at Yale, he had specialized in the study of electromagnetic waves, a phenomenon discovered by Heinrich Hertz in 1887. His subsequent work focused on the inadequacies of the Branly coherer, a device designed to detect these waves and a critical link in Marconi's system of wireless telegraphy. Despite research and invention that had spanned a number of years, De Forest's plans remained unrealized until a completely reliable detector was found to replace the Branly coherer.

The solution at last appeared, thanks to an observation Edison made in 1883 when he put a metal-wire electrode with a positive charge into his light bulb and found that electricity flowed from the glowing filament to the new electrode across the space between them. Twenty years later John Fleming discovered that the "Edison effect" bulb could detect hertzian waves in a completely new way. This was the first radio vacuum tube, the diode.

De Forest began experimenting with a simple detector that contained a gas flame instead of an electric filament. In addition, he used a telephone receiver—with a battery of its own—and clearly heard the dot-dash wireless signals from a distant transmitter. He added these two items to Fleming's diode and later tried a zigzag grid of wire between the filament and a metal-plate electrode to carry the incoming signal. De Forest named this first triode (patented in 1907) the "Audion" and much later called it "the granddaddy of all the vast progeny of electric tubes that have come into existence since."

But the progeny were not immediately forthcoming, partly because no one fully understood how the Audion really worked. Some gases remained in the tube's partial vacuum. To De Forest, it seemed that the current could only flow from filament to plate through a transporting medium—the ionized gases. It was an obvious explanation—but incorrect. By 1912 it had become clear that a transporting medium was unnecessary. The filament emitted electrons—particles with a negative charge—and the plate with its positive charge attracted them. High-vacuum tubes would be far more efficient than the "soft" gassy tubes used by De Forest until then.

De Forest also discovered that by feeding part of the output of his triode vacuum tube back into its grid, he could cause a self-regenerating oscillation in the circuit. The signal from this circuit was far more powerful and effective than that of the crude transmitters then generally employed. When appropriately modified, this single invention was capable of transmitting, receiving, or amplifying radio signals and it was not surpassed until the invention of the transistor in 1947.

31. According to the passage, the first triode could best be described as

(A) a rudimentary device that was later refined and improved
(B) a simple structure that functioned as efficiently as does a transistor
(C) a complex machine that ultimately proved too intricate to be practical
(D) an ill-conceived design that was quickly superseded by the work of other inventors
(E) a controversial apparatus that stunned scientists of the time

32. De Forest's statements quoted in the passage reveal which of the following attitudes?

(A) Impatience and modesty
(B) Idealism and detachment
(C) Determination and pride
(D) Fascination and awe
(E) Hostility and defiance

33. According to the passage, De Forest's explanation of how his original Audion functioned was incorrect because he misunderstood the

(A) role of electrons in ionized gases
(B) means by which electrons travel in a vacuum tube
(C) rate at which electricity is absorbed by the plate
(D) composition of the partial vacuum inside the tube
(E) necessity for both a filament and a plate

34. It can be inferred from the passage that De Forest's triode was capable of which of the following?

I. Intensification of the strength of the signal produced by a transmitter
II. Production of supplementary electrical power
III. Amplification of incoming signals in the receiver

(A) II only (B) III only (C) I and II only (D) I and III only (E) I, II, and III

35. Which of the following titles is most suitable for the passage?

(A) Founders of Modern Electronics
(B) Personalities of Early Radio
(C) The Discovery of Hertzian Waves
(D) How to Use the High-Vacuum Tube
(E) A Pioneer in Radio Technology

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Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. WOODS:TREES :: (A) lawn:grass (B) ocean:rocks (C) forest:paths (D) shore:lake (E) valley:stream

37. CROCODILE:REPTILE :: (A) frog:pond (B) bridge:gorge (C) tornado:storm (D) dawn:day (E) spring:flower

38. IMPOSE:BESTOW :: (A) stroll:walk (B) leave:depart (C) demand:ask (D) praise:honor (E) estimate:calculate

39. BOGUS:DECEIVE ::
(A) prudish:offend
(B) perilous:neglect
(C) listless:enliven
(D) mysterious:perplex
(E) threatening:safeguard

40. SHIRK:OBLIGATION ::
(A) desert:correction
(B) penalize:mistake
(C) recognize:authority
(D) default:payment
(E) blame:accusation

41. STRATUM:EARTH :: (A) seat:theater (B) story:building (C) sequence:event (D) trademark:company (E) stalk:plant

42. INTERLOPER:INTRUSION ::
(A) witness:interrogation
(B) actor:intermission
(C) recluse:interference
(D) mediator:intercession
(E) orator:interruption

43. PNEUMATIC:AIR :: (A) electric:light (B) insulated:cold (C) galvanized:iron (D) hydraulic:liquid (E) inorganic:solid

44. HABITABLE:DWELLERS ::
(A) literate:editors
(B) hospitable:combatants
(C) passable:travelers
(D) invulnerable:aggressors
(E) indestructible:developers

45. DIORAMA:SCENE :: (A) prologue:play (B) effigy:person (C) cartoon:movie (D) tableau:stage (E) drama:cast

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

| Correct Answers for Scholastic Aptitude Test Form Code 6G | | | |
|---|---|---|---|
| VERBAL | | MATHEMATICAL | |
| Section 2 | Section 6 | Section 5 | Section 1 |
| 1. D | 1. E | 1. D | 1. A |
| 2. D | 2. A | 2. D | 2. B |
| 3. B | 3. A | 3. C | 3. D |
| 4. B | 4. A | 4. D | 4. D |
| 5. C | 5. B | 5. E | 5. B |
| 6. E | 6. E | 6. C | 6. D |
| 7. B | 7. D | 7. A | 7. C |
| 8. A | 8. B | 8. B | *8. B |
| 9. A | 9. D | 9. C | *9. C |
| 10. B | 10. B | 10. E | *10. D |
| 11. C | 11. A | 11. D | *11. B |
| 12. B | 12. A | 12. A | *12. A |
| 13. E | 13. C | 13. E | *13. C |
| 14. E | 14. C | 14. A | *14. B |
| 15. C | 15. A | 15. A | *15. D |
| 16. D | 16. C | 16. C | *16. B |
| 17. A | 17. E | 17. E | *17. D |
| 18. C | 18. D | 18. C | *18. C |
| 19. C | 19. C | 19. D | *19. B |
| 20. B | 20. B | 20. C | *20. B |
| 21. B | 21. B | 21. B | *21. A |
| 22. B | 22. D | 22. C | *22. A |
| 23. C | 23. E | 23. E | *23. C |
| 24. B | 24. B | 24. D | *24. D |
| 25. C | 25. D | 25. B | *25. A |
| 26. A | 26. E | | *26. C |
| 27. C | 27. B | | *27. B |
| 28. E | 28. D | | 28. D |
| 29. E | 29. C | | 29. D |
| 30. B | 30. B | | 30. C |
| 31. D | 31. A | | 31. E |
| 32. D | 32. C | | 32. D |
| 33. B | 33. B | | 33. C |
| 34. E | 34. D | | 34. A |
| 35. E | 35. E | | 35. B |
| 36. C | 36. A | | |
| 37. E | 37. C | | |
| 38. C | 38. C | | |
| 39. D | 39. D | | |
| 40. E | 40. D | | |
| | 41. B | | |
| | 42. D | | |
| | 43. D | | |
| | 44. C | | |
| | 45. B | | |

*Indicates four-choice questions. (All of the other questions are five-choice.)

## The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.

- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

## How to Score the Test

### SAT-Verbal Sections 2 and 6

**Step A:** Count the number of correct answers for *section 2* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

**Step B:** Count the number of correct answers and the number of incorrect answers for *section 6* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

**Step C:** To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

**Step D:** To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

**Step E:** To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 298. Enter this figure on the worksheet.

### SAT-Mathematical Sections 5 and 1

**Step A:** Count the number of correct answers and the number of incorrect answers for *section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

**Step B:** Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 1* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

**Step C:** Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 1* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

**Step D:** To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

**Step E:** To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

**Step F:** To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 298. Enter this figure on the worksheet.

## SAT SCORING WORKSHEET

### SAT-Verbal Sections

A. Section 2: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 6: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on page 298.) [ ] SAT-verbal score

### SAT-Mathematical Sections

A. Section 5: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 1: Questions 1 through 7 and 28 through 35 (5-choice) _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 1: Questions 8 through 27 (4-choice) _____ no. correct − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on page 298.) [ ] SAT-math score

## Score Conversion Table
## Scholastic Aptitude Test
## Form Code 6G

| Raw Score | College Board Reported Score | | Raw Score | College Board Reported Score | |
|---|---|---|---|---|---|
| | SAT-Verbal | SAT-Math | | SAT-Verbal | SAT-Math |
| 85 | 800 | | 40 | 450 | 580 |
| 84 | 780 | | 39 | 450 | 570 |
| 83 | 760 | | 38 | 440 | 570 |
| 82 | 740 | | 37 | 430 | 560 |
| 81 | 730 | | 36 | 430 | 550 |
| 80 | 720 | | 35 | 420 | 540 |
| 79 | 710 | | 34 | 410 | 530 |
| 78 | 700 | | 33 | 410 | 520 |
| 77 | 690 | | 32 | 400 | 510 |
| 76 | 690 | | 31 | 390 | 500 |
| 75 | 680 | | 30 | 390 | 490 |
| 74 | 670 | | 29 | 380 | 480 |
| 73 | 660 | | 28 | 380 | 480 |
| 72 | 650 | | 27 | 370 | 470 |
| 71 | 650 | | 26 | 360 | 460 |
| 70 | 640 | | 25 | 360 | 450 |
| 69 | 630 | | 24 | 350 | 440 |
| 68 | 630 | | 23 | 340 | 430 |
| 67 | 620 | | 22 | 330 | 420 |
| 66 | 610 | | 21 | 330 | 420 |
| 65 | 600 | | 20 | 320 | 410 |
| 64 | 600 | | 19 | 310 | 400 |
| 63 | 590 | | 18 | 310 | 390 |
| 62 | 580 | | 17 | 300 | 390 |
| 61 | 580 | | 16 | 290 | 380 |
| 60 | 570 | 800 | 15 | 290 | 370 |
| 59 | 570 | 780 | 14 | 280 | 360 |
| 58 | 560 | 760 | 13 | 270 | 350 |
| 57 | 550 | 750 | 12 | 270 | 350 |
| 56 | 550 | 740 | 11 | 260 | 340 |
| 55 | 540 | 730 | 10 | 250 | 330 |
| 54 | 540 | 720 | 9 | 250 | 330 |
| 53 | 530 | 710 | 8 | 240 | 320 |
| 52 | 520 | 700 | 7 | 230 | 310 |
| 51 | 520 | 690 | 6 | 230 | 300 |
| 50 | 510 | 680 | 5 | 220 | 300 |
| 49 | 510 | 670 | 4 | 210 | 290 |
| 48 | 500 | 660 | 3 | 200 | 280 |
| 47 | 490 | 650 | 2 | 200 | 270 |
| 46 | 490 | 640 | 1 | 200 | 270 |
| 45 | 480 | 630 | 0 | 200 | 260 |
| 44 | 480 | 620 | –1 | 200 | 250 |
| 43 | 470 | 610 | –2 | 200 | 240 |
| 42 | 460 | 600 | –3 | 200 | 240 |
| 41 | 460 | 590 | –4 | 200 | 230 |
| | | | –5 | 200 | 220 |
| | | | –6 | 200 | 210 |
| | | | –7 | 200 | 210 |
| | | | –8 or below | 200 | 200 |